

An Agentic AI Search for Galaxy-Scale Strong Gravitational Lenses in the Public JWST NIRCам Archive

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Abstract

We report the first archive-wide search for galaxy-scale strong gravitational lenses conducted end-to-end by a general-purpose agentic AI. Using CLAUDE CODE (v2.1.224, model `claude-opus-5`), initiated by a single natural-language prompt, the agent designed and live-verified its own search pipeline, wrote and ran all code, and then acted as its own visual inspector, adversarial verifier, and literature auditor. From a census of 6,848 public JWST/NIRCам calibration-level-3 observations ($t_{\text{exp}} > 1000$ s; 916 distinct fields; 4.48 deg^2 of unique sky), the agent selected the complete population of 5,391 Legacy Surveys elliptical galaxies with $r < 21$, streamed $10''$ cutouts directly from the MAST S3 archive, and visually inspected 5,308 rendered cutouts (100%) with vision subagents, including 31 blind-injected COWLS lenses as positive controls. A three-persona adversarial verification stage (instrument-artifact, morphology, and lensing-geometry refuters) reduced 2,024 flagged cutouts to a ranked top-100, with blind Einstein-radius estimates and an independent literature crossmatch. The search re-discovered published lenses without prior knowledge — rank 1 is the double-source-plane lens SL2S J02176-0513 — while an independent audit finds 66 of the top 100 have no prior identification, including two grade-A galaxy-scale candidates and a group-scale system whose deflector is the brightest cluster galaxy of WH J135810.5-105636, with blind $\theta_E = 3.6'' \pm 0.4''$ implying $M(< 22 \text{ kpc}) \approx 3 \times 10^{12} M_\odot$. The campaign consumed ~ 76.5 M billable tokens across 1,281 subagents on a consumer subscription; the discovery pipeline itself ran in ~ 40 hours of wall-clock, roughly half of it waiting out usage-limit windows. The complete 2.2 GB corpus of agent transcripts is preserved, and we analyze it to characterize *how* the agent reasons about lensing evidence: a recall-biased screening pass that never asserted a confident false detection, refutation by named astrophysical mechanisms, and unprompted escalation from visual inspection to quantitative image analysis. We discuss completeness limits (48% recovery of injected controls; 54% of 239 catalogued lenses on inspected cutouts), calibration weaknesses, and the human-AI alignment loop this workflow enables.

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1 Introduction

Galaxy-scale strong gravitational lenses are precision instruments: they weigh galaxy dark-matter halos, magnify high-redshift sources, and, with time-variable sources, constrain H_0 . They are also

rare, of order one per $\sim 10^4$ massive ellipticals, so every generation of lens discovery has been an exercise in scaling human judgement. Visual searches of HST fields gave way to arc-finders, then to convolutional neural networks applied to wide-field surveys (Huang et al., 2020, 2021; Inchausti et al., 2025; Jacobs et al., 2019; Petrillo et al., 2017; Storfer et al., 2024), to systematic benchmarking (Metcalf et al., 2019), and to crowdsourcing (Garvin et al., 2022; Marshall et al., 2016). In the JWST era, the COSMOS-Web Lens Survey (COWLS) combined expert visual inspection over a single contiguous mosaic to find >400 candidates in 0.54 deg^2 (Mahler et al., 2025; Nightingale et al., 2025).

The public JWST archive as a whole, however, resists all of these approaches. It is not a survey: it is thousands of heterogeneous pointings in dozens of filter combinations, depths, and pipeline versions (Gardner et al., 2023; Rieke et al., 2023). No training set exists that matches its selection function, so supervised CNNs cannot be applied off the shelf; and its footprint is too scattered and too deep for a volunteer campaign to cover quickly. Meanwhile, large multimodal language models have reached the point where a single system can write and execute code against live astronomical services, read images, argue about morphology in astrophysical vocabulary, and search the literature — the full skill set of a lens-search team, if not yet its judgement.

This paper reports what happens when that experiment is actually run. We gave CLAUDE CODE (Anthropic, 2025), a general-purpose agentic AI operating a terminal on a consumer laptop, the following prompt (verbatim; see Appendix B):

“We want to try a one-shot galaxy scale strong lens search in the JWST archive. Limit this search to observations with exposure times greater than 1000 seconds that are publically available. Make cutouts that are 10 arc seconds wide centered on a galaxy that is brighter than 21st magnitude. Look at every cutout and determine whether it contains a strong lens. [...] For now, I would like to see the top 100 candidates.”

Everything downstream of that prompt — the archive census, the deflector catalog queries, the streaming cutout service, the decision to inject blind positive controls, the inspection prompts, the adversarial verification design, the ranking, and the literature crossmatch — was designed and executed by the agent, with the human operator supervising, approving the plan, and re-launching stages after usage-limit interruptions. To our knowledge this is the first strong-lens discovery campaign, and among the first astronomical discovery campaigns of any kind, run end-to-end by an agentic AI on a public archive.

Our contributions are:

1. **A complete archive-scale search** of public JWST NIRCам imaging around 5,391 bright elliptical deflectors — the complete $r < 21$ population in the covered footprint, not a sample — producing a ranked top-100 candidate list with per-candidate evidence text, blind Einstein-radius estimates, deflector redshifts, and prior-identification status (Appendix A).
2. **Blind validation inside the run:** 31 COWLS lenses (Nightingale et al., 2025) injected with randomized identifiers, plus an after-the-fact completeness measurement against 239 catalogued lenses that happen to fall on inspected cutouts.
3. **An independently audited novelty assessment:** 10 of the top 100 are published lenses (led by the double-source-plane system SL2S J02176-0513 at rank 1), 24 sit in known cluster strong-lensing fields, and 66 carry no prior identification.
4. **New candidates**, including two clean grade-A galaxy-scale systems and a group-scale arc system around the brightest cluster galaxy of the Wen and Han (2024) cluster WH J135810.5-105636 (Section 4.5).

5. **A reasoning corpus and its analysis:** 1,281 per-agent transcripts (2.2 GB) recording every image the system looked at, every tool call, and every verdict rationale. Section 5 mines this corpus to describe how the agent actually identifies lenses — including behaviors we did not prompt for.

Section 2 describes the archive census and target selection; Section 3 the agentic pipeline; Section 4 the candidate list, control recovery, and audit; Section 5 the transcript analysis; Section 6 compares this modality with CNN and citizen-science searches and lays out the human-alignment loop; Section 7 concludes. Throughout we use the [Planck Collaboration et al. \(2020\)](#) cosmology.

2 Data

2.1 JWST archive census

A single MAST `astroquery` census returned all public JWST/NIRCam calibration-level-3 `i2d` mosaics with $t_{\text{exp}} > 1000$ s: 6,848 observations. Footprint-overlap clustering (union-find on the `s_region` polygons) grouped them into 916 distinct sky fields covering 4.48 deg^2 of unique sky. No mosaics were downloaded: the agent verified during planning that the MAST public S3 mirror (`stpubdata`) serves uncompressed SCI extensions that support true HTTP ranged reads via `astropy.io.fits + fsspec`, so every cutout was streamed as a $\sim 10''$ section directly from the cloud. The entire run stayed under 1 GB of local disk — a constraint the agent discovered (a nearly full volume) and designed around.

2.2 Deflector selection

Deflectors are massive ellipticals from the DESI Legacy Imaging Surveys ([Dey et al., 2019](#)): DR10 (south) and DR9 (north), queried per field through the NOIRLab Data Lab TAP service. The selection is $r_{\text{dered}} < 21$ with Tractor type DEV, or SER with Sérsic index $n \geq 2.5$; a soft red-sequence cut $r - z > 0.4$ (rejections logged, not silently dropped); BRIGHT/MEDIUM maskbit exclusion; `shape_r` $< 8''$; $1''$ deduplication; and a tangent-plane point-in-polygon test against each observation’s actual footprint with a $6''$ inset. DR11 catalogs were confirmed to not be publicly served (the viewer exposes DR11 imagery only), so DR10/DR9 were used. A per-field cap of 60 targets (brightest first) preceded a global brightest-first trim. The final list contains 5,391 entries: 5,360 archive galaxies — the complete selected population in the covered footprint — plus 31 blind controls (Section 2.4).

Each target was assigned its best short-wavelength observation (preference order F150W, F200W, F115W, F090W, F210M, F150W2; deepest-exposure tiebreak) and best long-wavelength observation (F277W, F356W, F444W, F410M, F322W2).

2.3 Cutouts

For each target the agent rendered one JPEG with six panels in two rows: the full $10'' \times 10''$ field in (a) a shallow asinh stretch, (b) a deep asinh stretch, and (c) a two-band color composite (LW resampled onto the SW grid via WCS); and a central $3.5''$ zoom row showing (d) the deep stretch, (e) the color composite, and (f) a deflector-subtracted residual computed by radial profile subtraction. Panels carry a $1''$ scale bar, N/E compass, four yellow tick marks locating the catalogued galaxy, and corner quadrant labels (NE/NW/SE/SW). Of 5,391 targets, 5,308 rendered successfully and were inspected — 100% of renderable targets, with zero malformed verdicts — while 83 fell in NIRCam module gaps or had insufficient valid pixels.

2.4 Blind positive controls

The agent’s plan — adopted before any inspection began — injected 31 known and candidate lenses from the COWLS catalog (Mahler et al., 2025; Nightingale et al., 2025) into the target stream with randomized identifiers, indistinguishable from ordinary targets in the manifest. The control key was held aside and unblinded only at the reporting stage. The controls span COWLS ranking codes from unanimous grade-A to low-confidence candidates, with catalogued Einstein radii up to $\sim 2''$.

3 The agentic pipeline

3.1 Agent architecture and provenance

The search was run in CLAUDE CODE v2.1.224 with the `claude-opus-5` model (Anthropic, 2025, 2026) on a consumer subscription — these facts, along with per-message token usage and timestamps, are recorded on every line of the session transcripts, which makes the entire run auditable after the fact (Section 5). The architecture has three tiers: an interactive *orchestrator* session that talks to the operator, writes code, and launches work; deterministic *workflow* scripts that fan work out over parallel subagents with structured-output schemas; and the *subagents* themselves, each a fresh context that reads its assignment from a file, works with the full tool set (file reads, including images, shell, web), and returns validated JSON.

Three sessions comprise the campaign: two short *design* sessions and the main *discovery* session. The design sessions are notable for what they did before writing the plan: read-only subagents live-probed every external interface the plan would depend on — the S3 ranged-read path, the Data Lab TAP endpoints and their working ADQL dialect, PyPI wheel availability for the local Python, catalog-viewer fallbacks, disk headroom, and network reachability — so the plan document recorded *verified* facts rather than assumptions. The resulting `PLAN.md` fixed the seven-stage pipeline of Table 1, the exact selection criteria of Section 2.2, checkpointing for every stage, and the blind control injection, and was approved by the operator before execution.

Table 1: Pipeline stages with agent counts and billable tokens (input + output + cache-write), measured from the per-agent transcripts. Wall-clock spans include usage-limit stalls. The three verification personas each see every candidate assigned to them; the literature stage runs one agent per top-100 candidate plus catalog-acquisition agents.

Stage	Mechanism	Agents	Killed	Tokens	Wall span (UTC)
Census, catalogs, cutouts	Python scripts	—	—	—	Aug 8, 00:07–01:47
Plan & environment probes	design subagents	4	0	0.8M	Aug 8, 00:07–00:59
S5 vision inspection	13 workflows	537	90	27.4M	Aug 8, 01:47–21:23
S6 adversarial verification	12 workflows	526	326	22.0M	Aug 8–9
Literature crossmatch	8 workflows	192	51	23.0M	Aug 9–10
Catalog acquisition	2 workflows	11	0	2.1M	Aug 10
Candidate notes	3 workflows	11	0	1.2M	Aug 10
Total	39 workflows	1,281	467	76.5M	~3 days

3.2 Stage S5: vision inspection

Every rendered cutout was assigned, in deterministic shuffled batches of ~ 12 – 20 , to one of 537 vision subagents across 13 workflows. Each agent was instructed to **Read** every image in its batch —

the tool renders the JPEG into the model’s visual context — and to append one JSON verdict per cutout: `lens_at_center` (yes/likely/no), `quadrant_lens` (none/NE/NW/SE/SW), an `evidence` array, a `center_galaxy_type` classification, and a 0–100 confidence.

The inspection prompt (Appendix B) is a compact field guide: positive evidence (tangential arcs concave toward the deflector, multiple images at common radius and color, counter-images, blue-vs-red color contrast, features surviving in the residual panel) and named false-positive classes to rule out (spiral arms, collisional rings, edge-on disks, tidal debris, shells, diffraction spikes, chance alignments, and symmetric over-subtraction residuals). Critically, it declares the pass’s operating point: *“this is a high-recall first pass . . . the cost of a false alarm here is low, and the cost of a miss is total.”* Verdict files are append-only JSONL with resume-on-restart, which mattered: Section 3.5.

3.3 Stage S6: adversarial verification

All 2,024 flagged cutouts (yes or likely at center, or a quadrant flag) were scored, and the top 350 were passed to verification. Each candidate was re-examined by three independent subagent personas, each prompted to *refute* the lens claim from a different angle (Appendix B):

- **Instrument-artifact specialist:** diffraction spikes, snowballs, persistence, mosaic seams, striping, saturation bleed, and the symmetric over-/under-subtraction rings of the residual panel;
- **Galaxy-morphology specialist:** spiral arms, collisional and resonance rings, shells, tidal tails, mergers, edge-on disks, chance alignments;
- **Lensing-geometry specialist:** is the curvature concave toward the deflector with its center of curvature at the centroid, tangential elongation, counter-image or plausible image multiplicity, a plausible θ_E for the deflector’s luminosity, and color consistency among putative images of one source.

Each persona returns `pass/fail/uncertain` plus a named alternative explanation. The final grade is the pass count: A = 3, B = 2, C = 1, D = 0; flagged candidates outside the verified 350 are grade U. The ranking that produces the top 100 weights verifier grade above inspector confidence.

3.4 Blind θ_E measurement and literature crossmatch

Two further workflow families operated on the top 100 only. *Measurement* agents re-derived arc radii, position angles, counter-image assignments, and a geometric Einstein-radius estimate (arc radius, or half the image separation) from the imaging alone, with all published metadata withheld; these are the blind θ_E values of Table 2. *Literature* agents — one per candidate — converted coordinates, queried SIMBAD, NED, and VizieR, ran web searches on both coordinate forms, identified the JWST program, and in several cases fetched and read survey papers (`pdftotext` + string search) to decide between “published lens,” “lens-adjacent field,” and “no prior identification.” A separate catalog-acquisition family assembled ~195,000 catalogued lens and candidate entries (SIMBAD lens types, COWLS, the DESI-survey lens compilations of Huang et al. 2020, 2021; Inchausti et al. 2025; Storfer et al. 2024, and others) for uniform crossmatching.

3.5 Cost, interruptions, and robustness

Table 1 gives the accounting, measured directly from the per-message usage metadata in the 1,281 preserved transcripts: 76.5M billable tokens (8.5M output), 10,853 image reads, 5,053 shell

commands, and 2,466 web queries. The discovery pipeline itself (inspection + verification) cost 49.4M tokens — about 5,200 tokens per cutout inspected and $\sim 21,000$ per adversarial judgement — with the literature audit adding 26.3M more.

The dominant operational failure mode was the subscription’s usage limit: 467 of 1,281 subagent transcripts (36%) are stubs killed at launch by the limit, concentrated in the verification stage (326/526; the first verification tier lost 78–81% of its agents). The run’s own in-session accounting attributes $\sim 21\%$ of tokens to agents killed mid-flight, whose partial work was discarded. Because every stage checkpointed (append-only verdicts, skip-if-exists), the orchestrator re-launched only missing work after each reset: the final verification passes show zero kills and completed the grade assignments. Total wall-clock from prompt to delivered top-100 was ~ 40 hours, roughly half of it waiting out usage-limit windows; the literature audit ran over the following day.

4 Results

4.1 Verdicts, grades, and demographics

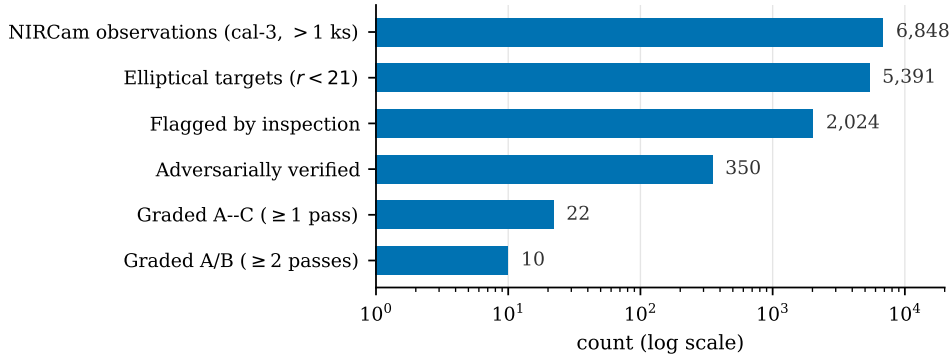


Figure 1: The selection funnel, from archive census to verified candidates. Every step after “Elliptical targets” was decided by the agent’s own inspection and verification verdicts.

Figure 1 summarizes the funnel. Of 5,308 inspected cutouts, the screening pass returned **yes** at center for 14, **likely** for 1,952, and a quadrant flag for 97; 2,024 unique cutouts entered the flagged pool with evidence text. The 350 verified candidates resolved into grades A: 5, B: 5, C: 12, D: 328. Central-galaxy classifications for the remainder are dominated by ellipticals (2,084), S0s (1,196), and edge-on disks (434), with mergers (291), spirals (196), stars (307), and artifacts (48) — the same confuser classes the verification stage was designed to kill.

4.2 Blind control recovery and completeness

The inspection stage flagged 15 of 31 blind controls (48%), and 3 of the 5 controls with unanimous COWLS grade-A rankings (Figure 3). Recovery shows no strong trend with Einstein radius in this small sample. No control reached the top 100: recovered controls came back at confidence 15–30 — inside the **likely** shortlist, below the verified tier.

The literature crossmatch enables a second, larger completeness measurement: 239 catalogued lenses and lens candidates happen to fall on inspected cutouts. The screening pass flagged 128 (54%) overall, and 22 of 31 (71%) of the SIMBAD-confirmed gLS systems. Of 26 known lenses that entered verification, the personas passed only 1 — a $\sim 92\%$ false-rejection rate on true lenses. The

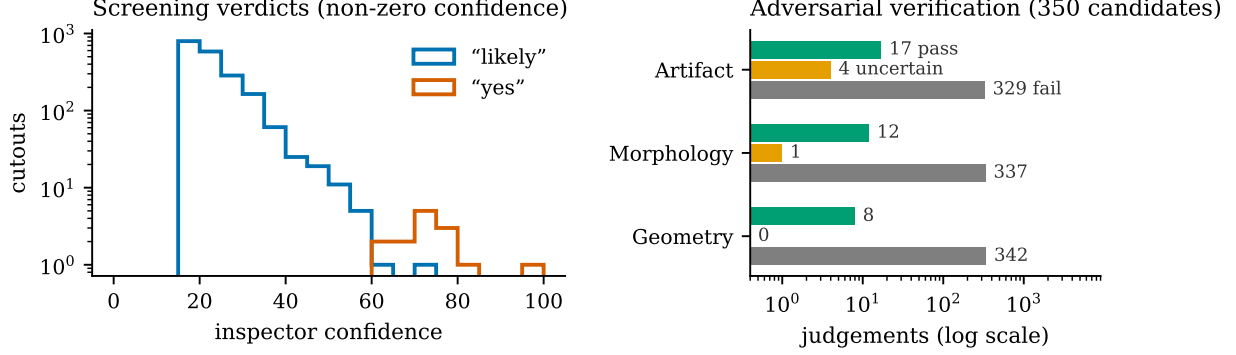


Figure 2: *Left:* inspector confidence for non-no verdicts. The screening pass behaves as instructed: **likely** means “worth a second look” (median confidence 20), and **yes** — median 72, maximum 95 — was asserted only 14 times in 5,308 cutouts. *Right:* verification outcomes per persona (log scale); of 350 candidates, each persona passed only 8–17 ($\sim 2\text{--}5\%$).

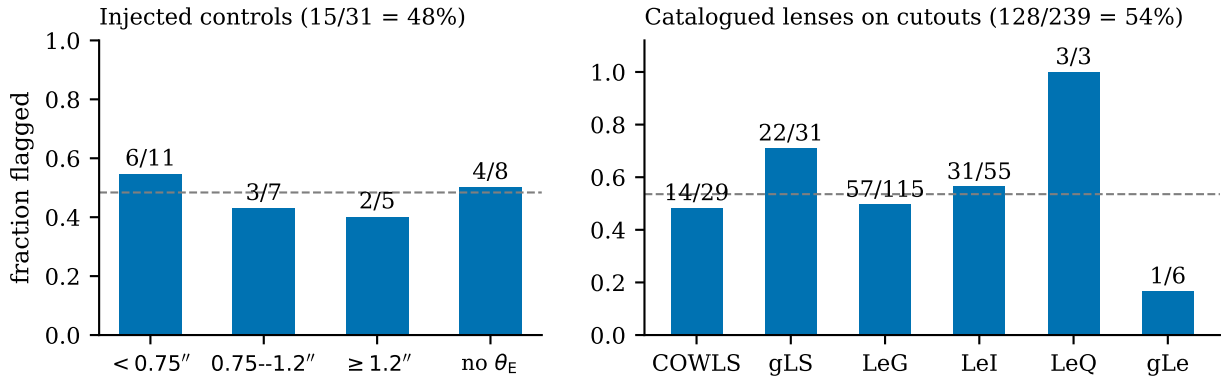


Figure 3: *Left:* recovery (flagged by inspection) of the 31 blind-injected COWLS controls, by catalogued Einstein radius. *Right:* after-the-fact recovery of the 239 catalogued lenses and candidates that fall on inspected cutouts, by catalog class (SIMBAD lens types; gLS = confirmed gravitational lens systems).

design conclusion is unambiguous and we state it as the run’s own README does: *a high grade is strong evidence for lensing; a low grade is not evidence against it*. The verification stage certifies; it does not grade recall. Completeness at COWLS-like depth is $\sim 50\%$, and the confirmed tier is a floor, not a census.

4.3 Independent novelty audit

The run’s own literature verdicts (Table 2, “Prior ID” column) report 7 published systems, 90 with no prior identification, and 3 judgement calls. Because novelty claims are the paper’s most fragile output, we re-ran the crossmatch *independently of the discovery session*, in a separate audited session against 43 catalogs (48,004 reference rows local + NED TAP + SIMBAD otypes + 19 vetted Vizier lens tables), with a dedicated cluster-lens layer that positional matching alone cannot provide (a cluster’s catalogued coordinate is its center, $1\text{--}2'$ from any given arc). The audit finds, out of 100:

- **10 published lenses** (grade A: 2; B: 2; C: 3; U: 3), all recovered blind by the pipeline. Rank 1 is SL2S J02176-0513 (Cabanac et al., 2007; Tu et al., 2009), the double-source-plane lens:

the deflector matches at $0.06''$, and the agent’s blind $\theta_E = 1.15'' \pm 0.20''$ and fold-configuration description match the published system (Figure 4). Rank 4 matches a published DESI-survey candidate (Inchausti et al., 2025); rank 6 a citizen-science HST candidate (Garvin et al., 2022); others include cluster-lensed images in Abell 1689, MACS J0416, and RCS2 J0327 fields.

- **24 in known cluster strong-lensing fields** (e.g., Abell 2537, Abell 370, Abell 2744, RELICS and SGAS clusters; Coe et al. 2019), where cataloged arcs exist within $\sim 1'$ but no specific published arc matches within $5''$. Ten of these sit in documented lensing clusters for which no machine-readable arc list could be found; those are *unresolved*, not confirmed new.
- **66 with no prior identification** anywhere the audit could reach. One caveat inherited by every such verdict: the SLED database is login-walled and was never queried.

The audit found zero false negatives in the run’s own “known” labels, but reclassified 24 of its “new” claims as cluster-field — including one grade-A and two grade-B systems. The surviving clean grade-A/B unknowns are ranks 2, 8, and 10, with the qualification that rank 8 lies in the core of the massive cluster MACS J1105.7–1014, whose convergence plausibly boosts a galaxy-scale deflector. This is the central methodological lesson of the audit: *for archive searches that revisit famous deep fields, the dominant re-discovery channel is cluster environments, and a galaxy-galaxy-style positional crossmatch alone will overstate novelty.*

4.4 New galaxy-scale candidates

The two cleanest new galaxy-scale candidates are rank 2 (Figure 5) and rank 10 (J23069956+2559453; grade A, $z_{\text{ph}} = 0.66$, blind $\theta_E = 1.37'' \pm 0.25''$), both surviving all three adversarial personas and carrying no prior identification in the audit. Rank 8 (J16644236-1024898; grade B, $z_{\text{ph}} = 0.41$) is a strong galaxy-scale configuration in the MACS J1105.7–1014 cluster core. We additionally note rank 3 (J34707505-219476; grade A) as a spectacular arc that the audit places $1.06'$ from the core of Abell 2537 with cataloged multiple images $29''$ away — almost certainly real lensing, but cluster-associated and requiring a cluster-aware model rather than a galaxy-galaxy interpretation. Spectroscopic confirmation of the arcs (none of the sources has a redshift) is the required next step for all of these.

4.5 A group-scale system: the BCG of WH J135810.5-105636

Rank 15 (Figure 6) is the system flagged during our own review as the most interesting single object in the list. The deflector (RA, Dec = $209.5438^\circ, -10.9433^\circ$; $r = 18.85$; $z_{\text{ph}} = 0.495 \pm 0.026$, Legacy Surveys) coincides to $< 0.1''$ with the brightest cluster galaxy of WH J135810.5-105636, cluster ID 712636 in the Wen and Han (2024) DESI Legacy Surveys cluster catalog at $z_{\text{cl}} = 0.481$ — a cluster catalogued from the same DECam imaging in which the arcs are too faint to have been noticed, and absent from every lens catalog the audit could reach. The measurement agents blindly traced a broad eastern arc (ridge radius $3.87''$) and a mirror-symmetric northwestern counter-arc ($3.36''$), both concave inward, giving $\theta_E = 3.6'' \pm 0.4''$ from half their separation.

Taking $z_l = 0.481$ and the blind θ_E , the Einstein radius is $R_E \simeq 22$ kpc and the enclosed projected mass is

$$M(< \theta_E) = \frac{c^2}{4G} \theta_E^2 \frac{D_l D_s}{D_{ls}} \approx (2.8\text{--}4.5) \times 10^{12} M_\odot \quad (z_s = 1\text{--}3), \quad (1)$$

corresponding to an SIS velocity dispersion of $\sim 420\text{--}530$ km s $^{-1}$ — group-to-cluster scale, consistent with the low-richness Wen & Han identification. The arc morphology (broad, clumpy, with both

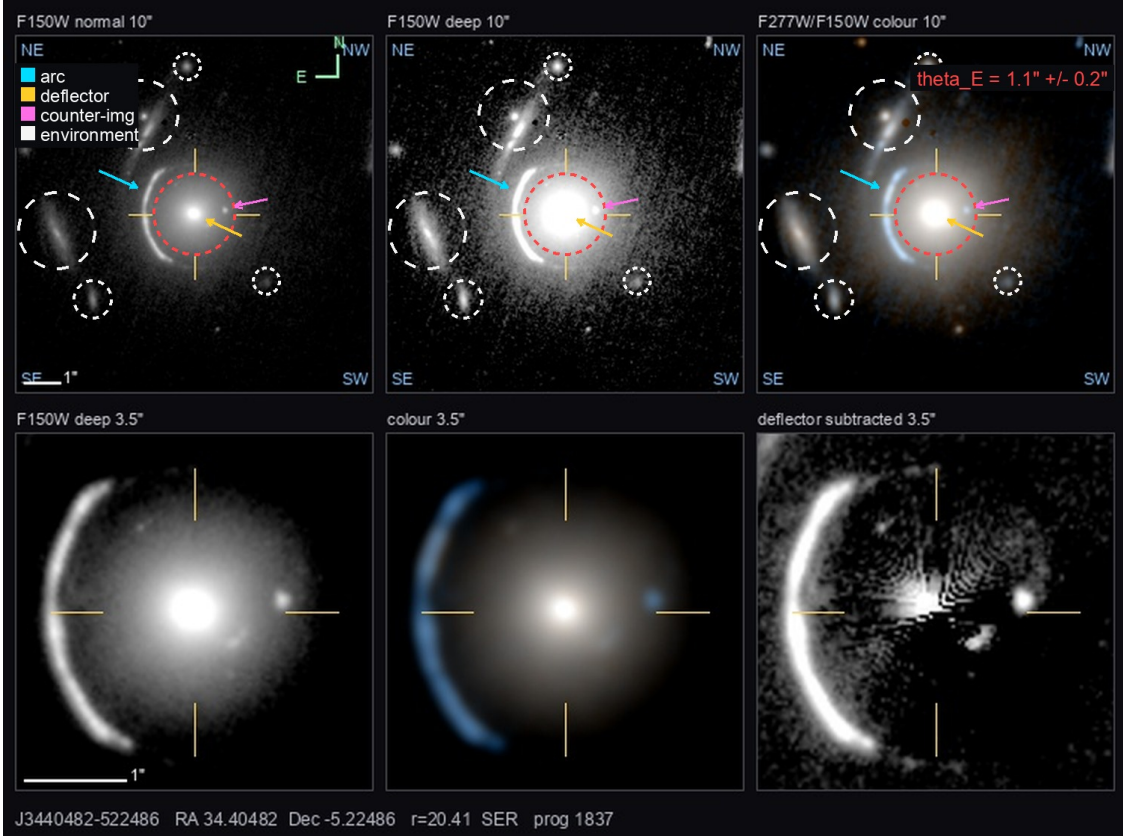


Figure 4: Rank 1, J3440482-522486 = SL2S J02176-0513, recovered blind at the top of the ranking — the validation anchor. Panels are the agent’s own annotated product: cyan = arc, yellow = deflector, magenta = counter-image, white = environment galaxies, red dashed = blind θ_E . Top row: 10'' field (normal, deep, color); bottom row: 3.5'' zoom (deep, color, deflector-subtracted).

blue knots and orange material) led the geometry verifier to withhold full endorsement (grade C), and its skepticism is recorded verbatim in the transcript: morphologically the debris interpretation was not excludable from the panels alone. A pixel-level lens model and source-redshift spectroscopy are the decisive follow-ups; both are planned with collaborators.

4.6 Data-quality caveats

The audit surfaced defects that we report as part of the result. Ranks 7 and 14 are the same object 1.17'' apart with different grades (B vs. C) — a deduplication miss across observations; ranks 16 and 17 are likely one system counted twice (8.78''). One candidate has a null deflector photo-z. The `candidate_id` strings mangle coordinate precision and must not be parsed as positions (the `ra_sex/dec_sex` columns are verified to 0.008''). Blind θ_E values are geometric estimates from rendered 8-bit panels, not lens models; all color statements in verdicts are composite-panel RGB ratios, not calibrated photometry. And every “no prior identification” verdict carries the un-queried-SLED caveat. None of these change the headline counts; all of them are exactly the kind of defect a human referee should expect an agentic pipeline to produce silently, which is why the independent audit exists.

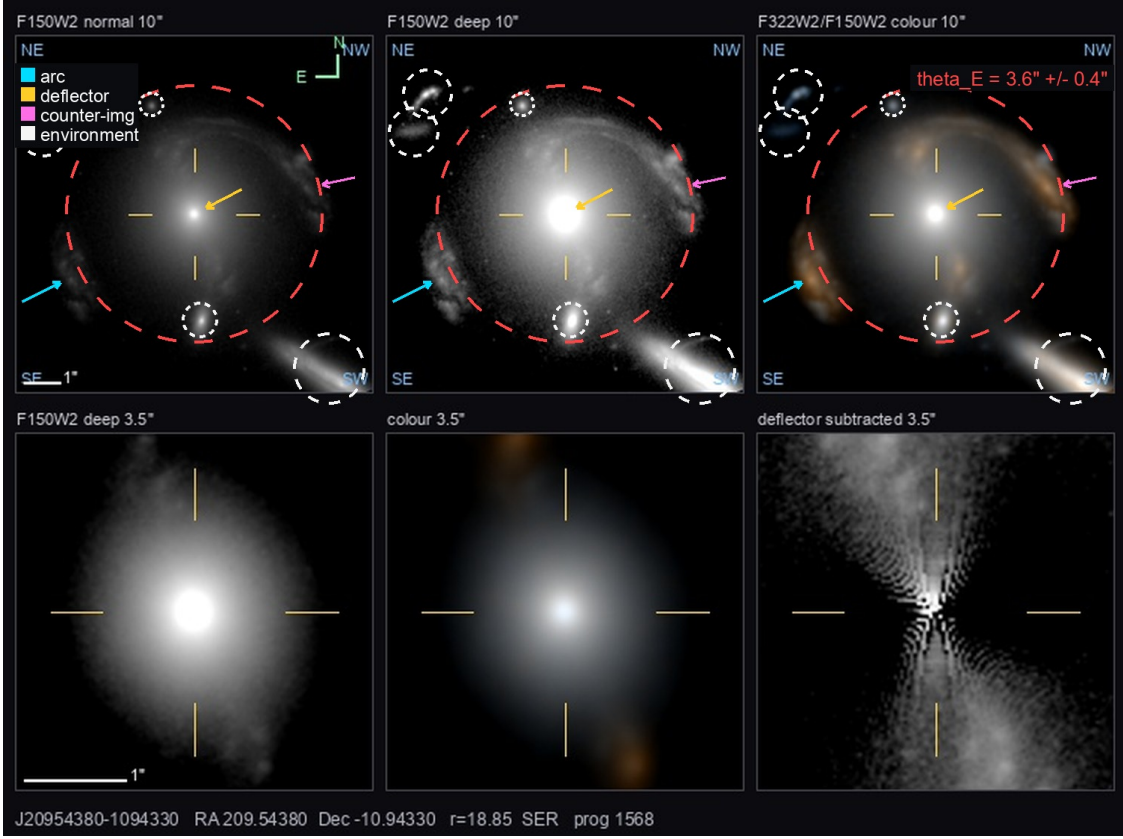


Figure 6: Rank 15, J20954380-1094330 (grade C): a $\theta_E = 3.6'' \pm 0.4''$ arc-plus-counter-arc system around the BCG of the Wen and Han (2024) cluster WH J135810.5-105636 ($z_{cl} = 0.481$), in the 24.5 ks F150W2/F322W2 imaging of JWST calibration program 1568. The mirror-symmetric arc pair (cyan/magenta) drives the blind θ_E ; the agent itself noted the northern thin arc is *not* concentric and excluded it from the estimate.

spikes, so a one-armed spiral / ring galaxy is not excluded”). Offsets are consistently quoted to $0.1''$ read off the panel scale bar. Across the flagged verdicts, the most frequent evidence terms are *residual* (1,103 of 2,024 verdicts), *knot* (1,039), *blue* (777), *arc* (549), *curved* (426), *ring* (384), *companion* (344), *concave* (305), and *counter-image* (244) — a vocabulary indistinguishable from a human lens-search team’s, deployed with positional quantification on every claim.

5.2 Calibration behavior: the shortlist, not the discovery

The screening prompt orders a high-recall pass, and the agents’ verdict distribution shows they internalized it to a degree we did not anticipate: in the largest inspection workflow (60 batches, 840 verdicts), the agents returned 523 no, 317 likely, and zero yes, with a self-imposed confidence ceiling of 50. Across the whole run, **yes** appeared 14 times in 5,308 cutouts (0.26%) — and the highest confidence of the entire campaign, 95, went to the cutout that is in fact SL2S J02176-0513. The screening stage, in other words, behaves as a shortlist generator that reserves assertion for overwhelming evidence, and the burden of proof is carried by the adversarial stage (Figure 2). The inverse is also true and is the pipeline’s key weakness: recovered known lenses came back at confidences 15–30, indistinguishable from the bulk of the shortlist (Section 4.2).

5.3 Verification argues by mechanism, and escalates to measurement

Verifier language is different in kind from inspector language: rather than describing features, it names generating mechanisms — “the classic bipolar model-mismatch butterfly,” “Sérsic-mismatch dipole,” “concentric-ripple over-subtraction artefact,” “spindle-shaped edge-on galaxy” — and kills candidates by radial-vs-tangential elongation, radius monotonicity, same-side geometry, or color identity with the host. The forced **alternative** field yields a machine-readable false-positive taxonomy over the 1,008 failed judgements: spiral arms and host rings (386), tidal/merger debris (193), chance alignments and companions (193), reduction and model artifacts (106), edge-on disks (60), and diffraction/PSF effects (16) — quantitatively matching the confuser classes an experienced human inspector lists first.

The geometry persona did something the prompts did not request: it wrote measurement code. Fifteen of fifteen final-pass geometry agents ran shell commands, and eleven performed genuine quantitative image analysis — one agent built a reusable script that converts panel pixels to arcseconds, masks the yellow astrometric ticks by RGB threshold, transforms to polar coordinates about the deflector, thresholds at the 92nd percentile, and fits a weighted algebraic circle to measure the arc radius, its radial scatter, and the offset of the center of curvature from the deflector centroid. A representative kill reads:

“Polar profiling about the nucleus shows the feature sweeps ~ 200 deg with its ridge radius growing monotonically from $1.43''$ at NW to $2.65''$ at SW, is $\sim 0.7''$ wide radially, ... and has no counterpart anywhere from E through S — a winding arm, not a constant-radius tangential arc.”
(geometry persona, fail; alternative: “inner ring / winding spiral arm in the host lenticular’s own disk”)

and a representative pass, for what became rank 10:

“A thin tangential arc runs SE-through-S at $r \sim 1.3''$ concave toward the core with its centre of curvature on the deflector centroid, it is measurably bluer than the halo at the same radius on the opposite side (R/B 1.13 vs 1.46) ... though no compact counter-image is resolved.”

Inspection agents, by contrast, stayed almost purely visual, escalating only as far as self-directed crop-and-zoom (31 of 99 agents in the sampled workflow re-rendered ambiguous panels at higher magnification before deciding). The escalation gradient — visual screening, mechanistic refutation, quantitative measurement — emerged from the personas’ goals, not from explicit instruction, and it concentrated the expensive analysis exactly where it mattered: candidates that survived it are the grade-A list. Persona disagreement is preserved too: rank 21, the published lens MACS0416-GGL1, passed morphology (“a smooth red DEV host ... has no disk or spiral structure capable of hosting star-forming knots, so my morphology attack cannot dispose of it”) while the artifact persona returned **uncertain** — the two verdicts, side by side, read like a referee exchange. Extended excerpts, with transcript file paths for every quote, are in Appendix C.

5.4 Literature agents and their discoveries about the tools

The per-candidate literature agents were the most tool-diverse: coordinate conversion in shell, web search on both coordinate forms, NED and SIMBAD cone searches, VizieR TAP, program identification, and — when a survey paper seemed relevant — fetching the PDF and grepping its text. Their verdicts draw exactly the distinction the audit later formalized: “that cluster IS a published cluster lens — but its published arc is $23.2''$ away and its BCG is $13.9''$ away, so this specific galaxy is not the published deflector.” The audit session’s own findings double as a field

report on the state of the infrastructure: `astroquery`’s NED module wraps a CGI endpoint that times out where NED’s TAP service answers in a second; SIMBAD’s primary `otype` misses lenses typed as radio galaxies unless the `otypes` table is queried; and naive VizieR matching against everything with “lens” in the title would have matched the RELICS *source* catalog and marked every candidate in those fields as known. Each of these is a silent false-negative or false-positive channel for any automated crossmatch, found because the agents’ failures were inspectable.

5.5 What the corpus does not show

The transcripts also delimit the modality’s current ceiling. No agent performed calibrated photometry or touched FITS pixels — all judgements, including the quantitative ones, operate on rendered 8-bit panels. No agent proposed a lens model. The redacted chain-of-thought means we characterize the reasoning by its externalized artifacts, not its internal process. And the duplicate-candidate defects of Section 4.6 passed through every stage unremarked: each agent sees only its assignment, and nothing in the architecture owns global consistency. These are not fundamental limits — FITS-native tooling, modeling subagents, and a consistency pass are all straightforward extensions — but they mark where human oversight was, and remains, load-bearing.

6 Discussion

6.1 Against the existing discovery modalities

The relevant comparison classes are CNN searches on uniform surveys and expert/citizen visual campaigns. Against CNNs (Huang et al., 2020, 2021; Inchausti et al., 2025; Jacobs et al., 2019; Petrillo et al., 2017; Storfer et al., 2024), the agentic search needed *no training set* — which is what made the heterogeneous JWST archive addressable at all — and it emits, for every candidate, an evidence paragraph, a named-alternative record, and a literature verdict that a human can adjudicate line by line. Its per-object cost ($\sim 5,200$ tokens to inspect, $\sim 21,000$ per verification judgement) is orders of magnitude above a CNN forward pass, and its throughput (5,308 cutouts in a working day, subscription-throttled) is orders of magnitude below a survey sweep; at DESI-survey scale (10^7 – 10^8 galaxies) this modality, today, is not a replacement but a *complement*: the natural division of labor is CNNs for uniform breadth and agents for heterogeneous archives, hard tails, and candidate adjudication. Against visual campaigns (Garvin et al., 2022; Mahler et al., 2025; Marshall et al., 2016), the agent’s 48–54% single-pass recall at COWLS-like depth is below expert performance on comparable imaging, but it is measured, repeatable, improvable by prompt-level alignment — and it took two days, not a campaign season.

Two further properties have no analog in either modality. First, *auditability*: the 2.2 GB transcript corpus is a complete causal record; every claim in the top-100 table can be traced to the images an agent read and the words it wrote. CNN saliency maps and volunteer click streams do not support this kind of retrospective cross-examination. Second, *closed-loop literature awareness*: the same system that found the candidates also determined, with documented queries, which were already known — a step that in conventional pipelines is a separate, human-driven project.

6.2 The honest failure modes

Our own expert review of the top 100, conducted on the contact sheet before the audit, identified four false-positive families: mergers, spiral/ring galaxies, image-reduction artifacts, and busy cluster fields. The verification stage’s own alternative-explanation taxonomy (Section 5.3) names the

same families, and the audit adds the fifth failure that neither stage owned: novelty overstatement in cluster fields. The completeness numbers are the sharpest limitation — a 92% verification false-rejection rate on known lenses means the grade system is a purity instrument only — and the weak confidence calibration means the ranking below the verified tier is “worth a look,” not a probability. All of these are measured, not anecdotal, which is what makes the next iteration tractable.

6.3 The human-alignment loop

The natural next experiment is already designed: split the graded top-100 (and the recovered-known set) into alignment and validation halves with matched lens counts; feed the alignment half’s expert annotations — which FPs fooled which persona, which real arcs the geometry checker under-weighted, which residual-panel artifacts recur — back into the inspection and verification prompts; and re-run on the validation half to measure the purity and completeness deltas. Because the “model” here is a prompt stack rather than trained weights, the alignment cycle costs hours, is fully documented in version control, and is legible to the domain experts driving it. A second extension, Phase B of the original plan, is JWST-native target selection (dropping the Legacy Surveys $r < 21$ elliptical prior) to reach high-redshift and spiral deflectors — the regime where the archive’s depth is unique and where COWLS (Nightingale et al., 2025) shows the yield lives.

6.4 Reproducibility

The run is re-executable from its repository (pipeline scripts, workflow definitions, prompts, manifests) against the same public archive, and its conclusions are re-derivable from the preserved verdict files without any model access. Exact re-generation of verdicts is not expected — the model samples, usage limits reshuffle batches, and the hosted model version will drift — so we treat reproducibility at the level that matters scientifically: the candidate list, the control-recovery numbers, and every quoted excerpt are checked against the archived artifacts, and the audit re-derived the novelty claims independently. Pinning open-weight models or logging seeds would tighten this further; transcript preservation is, for now, the strongest reproducibility instrument this modality has.

7 Conclusions

A single natural-language request — find strong lenses in the public JWST archive and show me the top 100 — was carried by a general-purpose agentic AI through pipeline design, live interface verification, code, census, streaming cutouts, exhaustive vision inspection with blind controls, three-persona adversarial verification, blind Einstein-radius measurement, literature crossmatching, and report generation, in about two days of wall-clock on a consumer subscription. The output is not a black-box score: it is a ranked, evidenced, literature-annotated candidate list with a complete audit trail.

Measured against that record: the pipeline re-found real lenses blind (SL2S J02176-0513 at rank 1 with the campaign’s highest confidence; 10 published systems in the top 100), delivered at least two clean new grade-A galaxy-scale candidates (ranks 2 and 10) and a group-scale arc system around a WH cluster BCG with $\theta_E = 3.6''$ (rank 15), recovered 48% of blind-injected COWLS controls and 54% of 239 catalogued lenses in its footprint, and overclaimed novelty in cluster fields until an independent audit corrected it. The transcript corpus shows a division of cognitive labor that no one prompted in detail: a screening pass that hedges, an adversarial pass that argues by

named mechanism and invents its own measurement code, and literature agents that distinguish a lensing cluster from a lensed galaxy.

Strong-lens discovery has scaled, in turn, by eyes, by crowds, and by convolutional filters. This experiment demonstrates a fourth modality — by argument: agents that look, measure, refute, and cite, leaving a record a referee can check. Its present completeness is modest and measured; its alignment loop is fast and human-legible; and the archive it opens — everything public, heterogeneous, and too irregular to train on — is precisely the data no other modality serves.

Software and data availability

The complete pipeline (scripts 01–20, workflow definitions with all prompts, and the session-level transcripts) and all result tables (`results.csv` for all 5,391 targets, the top-100 master table, control-recovery and known-lens-recovery tables, and the annotated candidate documents) are at <https://github.com/kvinneslandn-ML-AI/jwst-strong-lens-search>. The 1,281 per-agent transcripts (2.2 GB uncompressed) are archived alongside this paper’s analysis scripts in the project repository (`reproductions/agentic-lens-discovery/`); a Zenodo deposit is planned at submission. JWST data are from the public MAST archive (`stpubdata`); deflector catalogs and photometric redshifts from the DESI Legacy Imaging Surveys DR9/DR10 (Dey et al., 2019; Zhou et al., 2021); positive controls from COWLS (Nightingale et al., 2025).

Software: CLAUDE CODE v2.1.224 (Anthropic, 2025) running `claude-opus-5` (Anthropic, 2026); `astropy` (Astropy Collaboration et al., 2022), `astroquery`, `fsspec`, `numpy`, `scipy`, `matplotlib`, `pandas`, `Pillow`.

Facilities: JWST (NIRCam), MAST, Blanco/DECam (Legacy Surveys), Astro Data Lab, CDS (SIMBAD, VizieR), NED.

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A The top-100 candidate list

Figure 7 shows the full contact sheet; Table 2 lists all 100 candidates. Machine-readable versions, including the evidence and note text for every candidate, are in the data repository.

Table 2: The top-100 candidate list. θ_E is the blind geometric estimate (arc radius or half image-separation) measured from the imaging alone; grades: A = passed all three adversarial verifiers, B = two, C = one, U = flagged but not selected for verification. “Prior ID” is the run’s own literature verdict; the independent audit (Section 4.3) further reclassifies 24 rows as cluster-field matches. Exposure time is the SW-filter total.

#	ID	RA, Dec (J2000)	Grade	θ_E (″)	z_{ph}	z_{sp}	Filters	t_{exp} (ks)	Prior ID
1	J3440482-522486	02:17:37.156, -05:13:29.49	A	1.15 ± 0.20	0.697	0.6460	F150W/F277W	22.6	published

continued

#	ID	RA, Dec (J2000)	Grade	θ_E (")	z_{ph}	z_{sp}	Filters	t_{exp} (ks)	Prior ID
2	J15199556+2122210	10:07:58.936, +21:13:19.57	A	1.48±0.25	0.517	0.5094	F115W/F277W	7.5	none found
3	J34707505-219476	23:08:18.011, -02:11:41.13	A	1.40±0.30	0.305	–	F150W/F277W	1.2	none found
4	J30791374-4058431	20:31:39.299, -40:35:03.50	A	1.12±0.25	0.562	–	F150W/F277W	1.2	published
5	J34084513-960342	22:43:22.831, -09:36:12.31	B	1.35±0.20	0.443	–	F150W2/F322W2	1.8	none found
6	J18937456+6221686	12:37:29.894, +62:13:00.71	B	2.10±0.35	0.497	–	F200W/F444W	5.8	published
7	J18030075+2309921	12:01:12.180, +23:05:57.16	B	4.00±0.40	0.293	–	F150W2	1.8	none found
8	J16644236-1024898	11:05:46.167, -10:14:56.32	B	2.15±0.25	0.407	–	F150W2/F322W2	1.8	none found
9	J19789495-132348	13:11:34.789, -01:19:24.52	B	2.70±0.40	0.189	–	F150W/F277W	7.5	published
10	J23069956+2559453	15:22:47.895, +25:35:40.30	A	1.37±0.25	0.656	–	F115W/F277W	1.9	none found
11	J8999381-2012203	05:59:58.513, -20:07:19.32	C	2.00±0.30	0.261	–	F150W/F277W	3.3	none found
12	J2451534-2191925	01:38:03.683, -21:55:09.29	C	2.24±0.35	0.373	–	F150W/F277W	1.0	none found
13	J18805344+1121596	12:32:12.825, +11:12:57.45	C	0.72±0.20	0.518	0.5440	F150W/F277W	1.5	none found
14	J18030108+2309932	12:01:12.259, +23:05:57.56	C	4.60±0.35	0.266	–	F150W2	1.8	none found
15	J20954380-1094330	13:58:10.511, -10:56:35.88	C	3.60±0.40	0.495	–	F150W2/F322W2	24.5	none found
16	J5186648-1343587	03:27:27.956, -13:26:09.12	C	17.60±3.00	1.431	–	F150W2/F322W2	1.8	published
17	J5186803-1343778	03:27:28.328, -13:26:16.01	C	–	0.918	–	F150W2/F322W2	1.8	published
18	J4006002-161474	02:40:14.406, -01:36:53.07	C	0.81±0.20	0.595	–	F150W/F277W	10.3	none found
19	J33372744-1401097	22:14:54.585, -14:00:39.48	C	2.50±0.50	0.478	–	F150W/F277W	1.2	none found
20	J710586-7562357	00:28:25.406, -75:37:24.84	C	–	0.587	–	F150W2/F322W2	1.8	none found
21	J6403415-2406677	04:16:08.196, -24:04:00.39	C	1.20±0.30	0.455	–	F150W/F277W	37.1	published
22	J5414235-4704895	03:36:34.163, -47:02:56.20	C	1.70±0.40	0.655	–	F150W2/F322W2	1.8	none found
23	J18818168+1282960	12:32:43.604, +12:49:46.56	U	–	0.482	0.4659	F277W	–	none found
24	J6439169-1191885	04:17:34.005, -11:55:07.87	U	2.80±0.50	0.428	–	F150W/F277W	6.4	none found
25	J3996679-159719	02:39:52.029, -01:35:49.89	U	1.60±0.35	0.346	–	F150W/F277W	37.1	field
26	J357850-3044809	00:14:18.841, -30:26:53.11	U	2.45±0.40	0.323	–	F150W/F277W	21.3	none found
27	J15032484+248405	10:01:17.961, +02:29:02.59	U	0.83±0.25	0.345	0.3486	F150W/F277W	1.0	none found
28	J26831233+6515810	17:53:14.959, +65:09:29.17	U	1.18±0.25	0.577	–	F150W/F356W	33.6	none found
29	J20148636+4809579	13:25:56.726, +48:05:44.85	U	2.70±0.60	0.439	–	F444W	–	none found
30	J9006236-2015944	06:00:14.967, -20:09:34.00	U	2.50±0.40	0.934	–	F150W/F277W	5.0	none found
31	J26876212+6510625	17:55:02.910, +65:06:22.51	U	0.60±0.20	0.348	–	F150W/F356W	7.8	none found
32	J804181+1819597	00:32:10.034, +18:11:45.49	U	–	0.394	–	F150W/F277W	2.5	none found
33	J32232919-763201	21:29:19.006, -07:37:55.24	U	–	0.731	–	F150W/F277W	1.2	none found
34	J17771210-2809562	11:50:50.904, -28:05:44.23	U	–	0.402	–	F150W/F277W	2.5	possible
35	J3807110-4434755	02:32:17.064, -44:20:51.19	U	0.38±0.20	0.272	–	F150W/F277W	1.2	none found
36	J15055009+250185	10:02:12.021, +02:30:06.65	U	–	0.476	0.5799	F090W	1.2	none found
37	J15009907+202286	10:00:23.777, +02:01:22.31	U	3.30±0.45	0.422	0.4619	F150W/F277W	2.7	none found
38	J15010748+230270	10:00:25.796, +02:18:09.71	U	1.15±0.35	0.229	0.2199	F150W/F277W	22.0	none found
39	J16305610+3001292	10:52:13.464, +30:00:46.50	U	1.30±0.35	0.145	–	F150W/F277W	2.1	none found
40	J15029814+254087	10:01:11.554, +02:32:27.15	U	1.60±0.40	0.193	0.1116	F150W/F277W	1.0	none found
41	J34708849-217825	23:08:21.237, -02:10:41.71	U	0.95±0.30	0.252	–	F150W/F277W	1.2	none found
42	J12411163+1923209	08:16:26.791, +19:13:55.52	U	1.90±0.40	0.333	–	F150W/F277W	5.3	none found
43	J15049363+206401	10:01:58.471, +02:03:50.44	U	–	0.558	0.5469	F150W2/F444W	18.9	none found
44	J26807645+6520932	17:52:18.348, +65:12:33.55	U	0.98±0.30	0.591	–	F150W/F356W	33.6	none found
45	J8570140-2144889	05:42:48.335, -21:26:56.00	U	1.70±0.40	0.116	–	F444W	–	none found
46	J9002960-2014057	06:00:07.104, -20:08:26.04	U	1.10±0.30	0.418	–	F150W/F277W	5.0	none found
47	J15069997+222588	10:02:47.993, +02:13:33.15	U	–	0.316	–	F150W/F444W	2.2	none found
48	J4312165-2100739	02:52:29.196, -21:00:26.60	U	2.05±0.25	0.653	–	F150W2/F322W2	1.8	field
49	J5282421-774282	03:31:17.811, -07:44:34.15	U	–	0.380	–	F150W/F277W	1.0	none found
50	J20945435-1082161	13:57:49.043, -10:49:17.80	U	0.52±0.20	–	–	F322W2	–	none found
51	J4310830-518328	02:52:25.993, -05:10:59.81	U	–	0.316	–	F115W/F444W	2.2	none found
52	J14920654+6969086	09:56:49.570, +69:41:27.09	U	–	0.306	–	F200W/F444W	75.8	none found
53	J15067319+221999	10:02:41.566, +02:13:11.98	U	–	0.310	0.3054	F150W/F444W	2.2	none found
54	J4784995-5841089	03:11:23.988, -58:24:39.21	U	1.60±0.40	0.390	–	F200W/F444W	7.9	none found
55	J4205283+264369	02:48:12.679, +02:38:37.29	U	1.35±0.35	0.561	–	F150W2/F322W2	1.8	none found
56	J14977486+224978	09:59:05.966, +02:14:59.21	U	1.46±0.35	0.457	0.3895	F150W/F277W	1.0	none found
57	J21006317+1456150	14:00:15.160, +14:33:41.40	U	2.17±0.45	1.231	–	F150W/F322W2	113.4	none found
58	J2235243-7358297	01:29:24.583, -73:34:58.70	U	–	0.450	–	F150W/F277W	1.3	none found
59	J3419795-502846	02:16:47.507, -05:01:42.45	U	1.17±0.40	0.257	–	F444W	–	none found
60	J20960149-304690	13:58:24.358, -03:02:48.85	U	1.50±0.35	0.273	–	F277W	–	none found
61	J8149689-2410706	05:25:59.254, -24:06:25.43	U	–	0.714	–	F200W/F356W	2.8	none found
62	J15261689-1266966	10:10:28.052, -12:40:10.77	U	1.25±0.30	0.517	–	F150W2/F322W2	1.8	none found
63	J15067092+210603	10:02:41.021, +02:06:21.72	U	1.65±0.35	0.329	0.3392	F150W2/F444W	18.9	none found
64	J20865088+7724666	13:54:36.212, +77:14:47.96	U	1.25±0.40	0.410	–	F150W2/F322W2	1.8	none found
65	J4429240-2347200	02:57:10.177, -23:28:19.21	U	1.55±0.40	0.537	–	F150W/F277W	1.2	none found

continued

#	ID	RA, Dec (J2000)	Grade	θ_E (")	z_{ph}	z_{sp}	Filters	t_{exp} (ks)	Prior ID
66	J18850875+1133833	12:34:02.101, +11:20:17.98	U	1.28±0.35	0.499	–	F277W	–	none found
67	J20948181-1096952	13:57:55.633, -10:58:10.27	U	–	0.358	–	F322W2	–	none found
68	J17772978-2808033	11:50:55.148, -28:04:49.20	U	2.46±0.40	0.387	–	F150W/F277W	2.5	none found
69	J18931633+6219972	12:37:15.919, +62:11:59.00	U	0.89±0.35	0.108	–	F150W/F277W	5.7	none found
70	J17174762+4246955	11:26:59.428, +42:28:10.39	U	0.66±0.25	0.351	–	F150W/F277W	1.9	none found
71	J12391133+1822895	08:15:38.720, +18:13:44.23	U	1.66±0.45	0.533	–	F150W2/F322W2	8.6	none found
72	J21007480+1455269	14:00:17.951, +14:33:09.70	U	1.31±0.30	0.480	–	F150W/F322W2	113.4	none found
73	J8575205-2143155	05:43:00.492, -21:25:53.57	U	1.10±0.35	0.450	–	F444W	–	none found
74	J16193787+1252943	10:47:45.088, +12:31:45.96	U	1.55±0.40	0.459	–	F277W	–	none found
75	J35084320-3043955	23:23:22.368, -30:26:22.40	U	0.73±0.30	0.342	–	F200W/F322W2	5.0	none found
76	J8732838-6207334	05:49:18.810, -62:04:24.03	U	2.50±0.40	0.408	–	F150W2/F322W2	1.8	none found
77	J9192602-4479623	06:07:42.246, -44:47:46.44	U	1.45±0.35	0.232	–	F150W2/F322W2	1.8	none found
78	J18895415+499333	12:35:48.996, +04:59:35.98	U	–	0.592	–	F150W/F444W	74.7	none found
79	J16456132+2953030	10:58:14.717, +29:31:49.06	U	2.96±0.60	0.494	–	F200W/F356W	2.8	none found
80	J15424410+3903685	10:16:58.583, +39:02:12.65	U	1.39±0.35	0.653	–	F150W2/F322W2	1.8	none found
81	J31352710-005281	20:54:06.504, -00:03:10.11	U	–	0.258	–	F150W/F277W	7.5	none found
82	J21483295+5282544	14:19:19.907, +52:49:31.57	U	2.30±0.40	0.387	0.3537	F150W/F277W	2.8	none found
83	J3470183+013617	02:18:48.440, +00:08:10.23	U	–	0.205	–	F200W/F356W	2.8	none found
84	J15057245+223798	10:02:17.387, +02:14:16.73	U	–	0.102	0.1248	F150W/F444W	2.2	none found
85	J20861957+7726170	13:54:28.697, +77:15:42.12	U	2.00±0.35	0.419	–	F150W2/F322W2	1.8	none found
86	J21518078+5278672	14:20:43.388, +52:47:12.19	U	–	0.444	–	F115W/F277W	2.8	none found
87	J26816024+6505016	17:52:38.458, +65:03:00.59	U	0.82±0.30	0.640	–	F200W/F356W	31.1	none found
88	J32840780+1769774	21:53:37.871, +17:41:51.85	U	1.45±0.35	0.272	–	F150W2/F322W2	1.8	none found
89	J9003772-2015513	06:00:09.053, -20:09:18.48	U	1.15±0.35	0.364	–	F150W/F277W	3.3	none found
90	J1637643+1340558	01:05:30.343, +13:24:20.10	U	2.70±0.45	0.216	–	F150W2/F322W2	1.8	none found
91	J14995729+243834	09:59:49.750, +02:26:18.02	U	2.00±0.40	0.365	0.3596	F150W/F277W	1.0	none found
92	J2104758+067717	01:24:11.418, +00:40:37.80	U	0.90±0.20	0.233	–	F200W/F444W	1.1	none found
93	J11084396-7342226	07:23:22.550, -73:25:20.12	U	–	0.247	–	F115W/F444W	39.5	none found
94	J4637336-3182658	03:05:29.607, -31:49:35.69	U	–	0.365	–	F200W/F356W	2.8	none found
95	J34135498+2813722	22:45:25.196, +28:08:13.99	U	–	0.373	–	F150W2/F322W2	1.8	none found
96	J33389992-3276033	22:15:35.980, -32:45:37.19	U	1.95±0.35	0.722	–	F150W2/F322W2	1.8	none found
97	J18939200+6216064	12:37:34.079, +62:09:38.30	U	1.30±0.35	0.349	–	F200W/F444W	5.8	none found
98	J1747804-3079539	01:09:54.730, -30:47:43.40	U	0.82±0.25	0.254	–	F200W/F356W	2.8	none found
99	J35128272-4120137	23:25:07.853, -41:12:04.95	U	0.69±0.35	0.378	–	F150W/F277W	1.2	none found
100	J6928514+070232	04:37:08.434, +00:42:08.37	U	2.24±0.30	0.291	–	F200W/F277W	4.1	none found

B Prompts and schemas

The initiating prompt (operator, verbatim).

“We want to try a one-shot galaxy scale strong lens search in the JWST archive. Limit this search to observations with exposure times greater than 1000 seconds that are publically available.

Make cutouts that are 10 arc seconds wide centered on a galaxy that is brighter than 21st magnitude.

Look at every cutout and determine whether it contains a strong lens. Tell me if there is a strong lens at the center of the cutout. If not, tell me if there is a strong lens in one of the quadrants and if so WHICH quadrant.

For a cutout that contains a strong lens, give me the evidence that this judgement is based on. Otherwise, tell me what type of galaxy is at the center of the cutout.

For now, I would like to see the top 100 candidates.”

A follow-up message during planning set the target count: “lets change the number of cutouts to 5000 instead of 10000.”

Inspection prompt (excerpt). Each inspection subagent received the panel-layout description, the evidence field guide (tangential arcs, multiple images, Einstein rings, counter-images, color contrast, residual persistence), the false-positive exclusion list (spiral arms, collisional rings, edge-on



Figure 7: All 100 ranked candidates, one three-panel strip each (normal stretch, deep stretch, color composite). Border color encodes the verifier tier (green: 3/3; blue: 2/3; thin blue: 1/3; gray: flagged, unverified); amber matting marks systems the crossmatch found already published. A full-resolution version is in the data repository.

disks, tidal tails and mergers, shells, diffraction spikes, chance alignments, symmetric over-subtraction residuals), and the operating point:

“CALIBRATION — THIS IS A HIGH-RECALL FIRST PASS. Three independent adversarial verifiers re-examine everything you flag and remove false positives. So the cost of a false alarm here is low, and the cost of a miss is total: a lens you mark ‘no’ is never looked at again. ... Use ‘likely’ generously. It does NOT mean ‘probably a lens’, it means ‘worth a second look’.”

The verdict schema per cutout:

`id; lens_at_center (yes/likely/no); quadrant_lens (none/NE/NW/SE/SW); evidence (array); center_galaxy_type; confidence (0–100).`

Verifier personas (excerpts). Three refuter briefs, one per persona, each ending in a named-alternative requirement:

Artifact: “You are an INSTRUMENTAL ARTIFACT specialist. Your job is to REFUTE each claim by finding a non-astrophysical explanation. Look specifically for: JWST 6-spike diffraction pattern ... snowballs, cosmic-ray showers, wisps, persistence; mosaic seams ... over- or under-subtraction rings in the residual panel: these are SYMMETRIC and centred on the galaxy, unlike a genuine offset arc ...”

Morphology: “... show the feature belongs to the galaxy itself or to an unrelated neighbour: spiral arms attached to the nucleus ... collisional / resonance ring galaxies ... shell ellipticals ... A genuine lensed arc is bluer than the deflector, has no stellar continuum link to the nucleus, and curves concave toward the deflector centre.”

Geometry: “... is the arc curvature CONCAVE TOWARD the candidate deflector, and is the centre of curvature near the deflector centroid? Is the feature tangentially elongated? ... Is the implied Einstein radius plausible (roughly 0.3–3'' for a galaxy-scale deflector; larger implies a group/cluster)? Is the deflector luminous/massive enough ...”

C Extended transcript excerpts

Each excerpt below is quoted verbatim from the named per-agent transcript (paths relative to the transcript archive; all verified against the manifest). They are chosen to show one complete example of each reasoning mode discussed in Section 5.

A routine inspection verdict pair (`wf_129a5e5a-bc8/agent-a50be3f89c21e2d06`): *“blue compact knot $\sim 2.3''$ NW of the red central galaxy”; “no counter-image seen opposite, so probably an unrelated blue source” (S0, confidence 18); “compact residual knot $\sim 1.2''$ SW of centre in the deflector-subtracted panel”; “asymmetric arc-like residual on the E side of the centre” (elliptical, confidence 25).*

A hedged flag with named null hypothesis (`wf_159c2096-f41/agent-af13aed0b3fb3c2d8`, confidence 45): *“prominent C-shaped crescent ~ 1.2 – $1.5''$ E of centre, concave toward the deflector, spanning NE through SE / crescent is bluish-white against an orange point-like nucleus (colour contrast) / arc persists in the deflector-subtracted panel ... / caveat: nucleus shows strong 6-ray JWST diffraction spikes, so a one-armed spiral / ring galaxy is not excluded.”*

Quantitative refutation, geometry persona (wf_864202f5-82c/agent-a2a5bc96023c91993, fail): the agent built an azimuthal-median model of the host, subtracted it, printed ridge radius versus position angle in 10° bins, and concluded: “*Polar profiling about the nucleus shows the feature sweeps ~ 200 deg (SW through W to NE) with its ridge radius growing monotonically from $1.43''$ at NW to $2.65''$ at SW, is $\sim 0.7''$ wide radially, is only a $\sim 20\%$ shoulder on the galaxy profile, and has no counterpart anywhere from E through S — a winding arm, not a constant-radius tangential arc.*” Alternative: “*inner ring / winding spiral arm in the host lenticular’s own disk.*”

A pass under attack, with persona disagreement (morphology persona, wf_70b18f44-c8f/agent-a00bb8b4c5b97a2e9, on the published lens MACS0416-GGL1, rank 21): “*Claimed evidence understates it — the deep zoom shows a chain of blue clumps at a nearly constant $r = 1.00\text{--}1.16''$ arcing from NE through N to NNW, concave toward a smooth red DEV host that has no disk or spiral structure capable of hosting star-forming knots, so my morphology attack cannot dispose of it.*” The artifact persona (wf_f61affa8-59b/agent-a8575beb8c2503273) returned uncertain on the same object.

Literature adjudication (wf_037410c8-5da/agent-a05248ab163bf08cc, rank 1): “*SIMBAD returns the deflector SDSS J021737.16-051329.5 ... at 0.14 arcsec, and the lens-system entry SL2S J021737-051329 (type gLS) at 0.98 arcsec. NED returns ‘SL2S J0217-0513 FG’ ($z = 0.6462$) at 0.06 arcsec.*” And the cluster-field distinction (agent-a6d48a517c1cc5575, rank 8): “*The position falls in the core of the known massive cluster MACS J1105.7-1014 ($z = 0.466$) ... and that cluster IS a published cluster lens — but its published arc is $23.2''$ away and its BCG ... is $13.9''$ away, so this specific galaxy is not the published deflector.*”

References

- Anthropic. Claude Code: an agentic command-line tool for software engineering. <https://claude.com/claude-code>, 2025. Version 2.1.224 used in this work.
- Anthropic. Claude Opus model family: system card. <https://www.anthropic.com>, 2026. Model claude-opus-5; TODO link the specific system card.
- Astropy Collaboration, A. M. Price-Whelan, P. L. Lim, N. Earl, N. Starkman, L. Bradley, et al. The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *The Astrophysical Journal*, 935(2):167, 2022. doi: 10.3847/1538-4357/ac7c74.
- R. A. Cabanac, C. Alard, M. Dantel-Fort, B. Fort, R. Gavazzi, P. Gomez, J.-P. Kneib, O. Le Fèvre, et al. The CFHTLS strong lensing legacy survey. I. Survey overview and T0002 release sample. *Astronomy & Astrophysics*, 461(3):813–821, 2007. doi: 10.1051/0004-6361:20065810.
- D. Coe, B. Salmon, M. Bradač, L. D. Bradley, K. Sharon, A. Zitrin, A. Acebron, C. Cerny, et al. RELICS: Reionization Lensing Cluster Survey. *The Astrophysical Journal*, 884(1):85, 2019. doi: 10.3847/1538-4357/ab412b.
- A. Dey, D. J. Schlegel, D. Lang, R. Blum, K. Burleigh, X. Fan, J. R. Findlay, D. Finkbeiner, D. Herrera, et al. Overview of the DESI Legacy Imaging Surveys. *The Astronomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

- J. P. Gardner, J. C. Mather, R. Abbott, J. S. Abell, M. Abernathy, F. E. Abney, J. G. Abraham, et al. The James Webb Space Telescope Mission. *Publications of the Astronomical Society of the Pacific*, 135(1048):068001, 2023. doi: 10.1088/1538-3873/acd1b5.
- E. O. Garvin, S. Kruk, C. Cornen, R. Bhatawdekar, R. Kannan, and B. Merin. Hubble Asteroid Hunter II. Identifying strong gravitational lenses in HST images with crowdsourcing. *Astronomy & Astrophysics*, 667:A141, 2022. doi: 10.1051/0004-6361/202243745.
- X. Huang, C. Storfer, V. Ravi, A. Pilon, M. Domingo, D. J. Schlegel, S. Bailey, A. Dey, R. R. Gupta, D. Herrera, et al. Finding Strong Gravitational Lenses in the DESI DECam Legacy Survey. *The Astrophysical Journal*, 894(1):78, 2020. doi: 10.3847/1538-4357/ab7ffb.
- X. Huang, C. Storfer, A. Gu, V. Ravi, A. Pilon, W. Sheu, R. Venguswamy, S. Banka, A. Dey, M. Landriau, et al. Discovering New Strong Gravitational Lenses in the DESI Legacy Imaging Surveys. *The Astrophysical Journal*, 909(1):27, 2021. doi: 10.3847/1538-4357/abd62b.
- I. Inchausti, X. Huang, et al. Strong Lens Discovery in the DESI Legacy Imaging Surveys Data Release 10 with Two Neural-Network Architectures. *arXiv e-prints*, 2025. TODO verify author list and journal status.
- C. Jacobs, T. Collett, K. Glazebrook, E. Buckley-Geer, H. T. Diehl, H. Lin, T. McKay, A. More, et al. An Extended Catalog of Galaxy–Galaxy Strong Gravitational Lenses Discovered in DES Using Convolutional Neural Networks. *The Astrophysical Journal Supplement Series*, 243(1):17, 2019. doi: 10.3847/1538-4365/ab26b6.
- G. Mahler, J. W. Nightingale, R. Massey, et al. The COSMOS-Web Lens Survey (COWLS) II: depth, resolution, and NIR coverage from JWST reveal 17 spectacular lenses. *Monthly Notices of the Royal Astronomical Society*, 544(1):L8, 2025. doi: 10.1093/mnras/slaf055.
- P. J. Marshall, A. Verma, A. More, C. P. Davis, S. More, A. Kapadia, M. Parrish, C. Snyder, et al. SPACE WARPS – I. Crowdsourcing the discovery of gravitational lenses. *Monthly Notices of the Royal Astronomical Society*, 455(2):1171–1190, 2016. doi: 10.1093/mnras/stv2009.
- R. B. Metcalf, M. Meneghetti, C. Avestruz, F. Bellagamba, C. R. Bom, E. Bertin, R. Cabanac, et al. The strong gravitational lens finding challenge. *Astronomy & Astrophysics*, 625:A119, 2019. doi: 10.1051/0004-6361/201832797.
- J. W. Nightingale, G. Mahler, N. B. Hogg, A. Amvrosiadis, Q. He, R. Massey, et al. The COSMOS-Web Lens Survey (COWLS) I: discovery of >100 high redshift strong lenses in contiguous JWST imaging. *Monthly Notices of the Royal Astronomical Society*, 543(1):203, 2025. doi: 10.1093/mnras/staf1522.
- C. E. Petrillo, C. Tortora, S. Chatterjee, G. Vernardos, L. V. E. Koopmans, G. Verdoes Kleijn, N. R. Napolitano, G. Covone, P. Schneider, A. Grado, and J. McFarland. Finding strong gravitational lenses in the Kilo Degree Survey with Convolutional Neural Networks. *Monthly Notices of the Royal Astronomical Society*, 472(1):1129–1150, 2017. doi: 10.1093/mnras/stx2052.
- Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Baccigalupi, M. Ballardini, et al. Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

